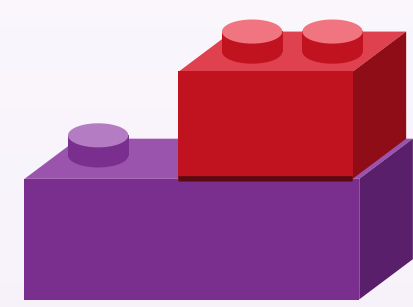




LEGO: Synergizing Expert GraphRAG and Expert Chain-of-Thought for Legal Reasoning

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Code & data
github.com/BLK-WHT/LEGO

Textual similarity $\neq$ legal relevance

Legal knowledge is **semantically sparse**. Distant provisions are linked through prerequisites, exceptions and general-special relations, while similar-looking ones can mislead.

Two bottlenecks

- **Retrieval.** RAG and GraphRAG rank by similarity and miss the normative links between provisions.
- **Reasoning.** Vanilla CoT can skip an exception, assume a prerequisite that is not met, or reason backwards from effect to condition.

Case in the figure. "Transport" appears 5 times, pulling models toward the carrier (**D**). But B has no contract with the carrier: privity (**Art. 465**) governs carrier liability (**Art. 832**), and **Art. 593** blocks bypassing it. So B claims from **A**.

Contributions & take-away

Adding expert legal structure to both retrieval and reasoning lets an 8B model match much larger LLMs on multi-hop civil-law questions.

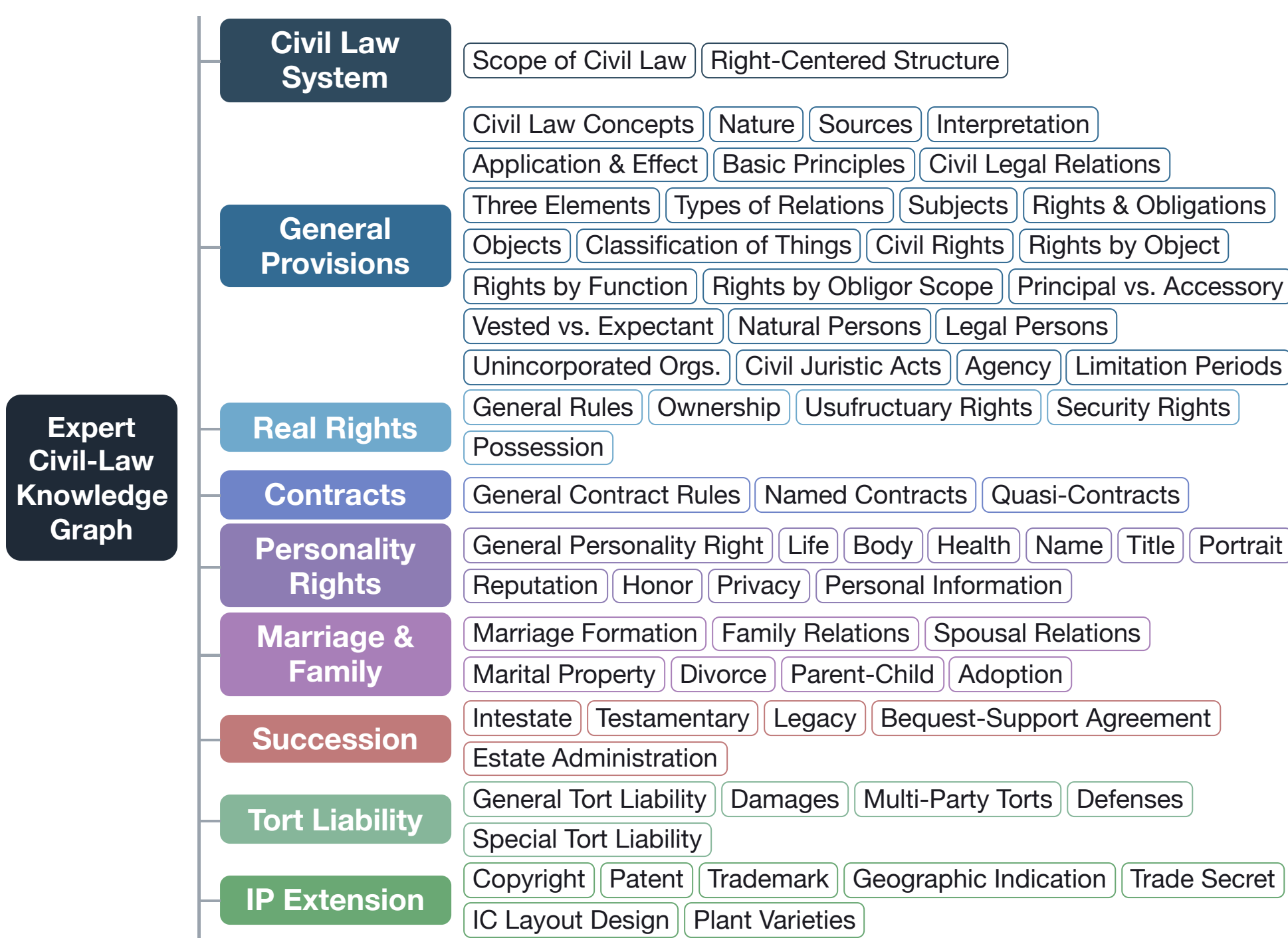
- **ExpertGraphRAG:** coverage-aware retrieval over an expert Civil Code graph
- **ExpertCoT:** Provision–Fact–Conclusion syllogistic reasoning with an auditable rationale
- Best accuracy on LawExamQA_Civil, stable up to ≥ 4 hops, and best on two open-ended benchmarks

ExpertGraphRAG

Retrieve by normative coverage, not similarity alone

Civil Law ExpertGraph

318 concepts **409** rule cards **4,152** support edges **1,211** of 1,260 articles



Normative Coverage Greedy (NCG)

Instead of similarity top-k, NCG adds an article only if it covers a legal rule the current set does not yet cover.

$$\Delta(a|S) = d(a) + \lambda_{\text{align}} g(a) + \lambda_{\text{cov}} |\Gamma(a) \setminus \Gamma(S)| - \lambda_{\text{red}} |\Gamma(a) \cap \Gamma(S)|$$

$d(a)$ relevance: query–article similarity

$g(a)$ expert alignment through rule cards

$|\Gamma(a) \setminus \Gamma(S)|$ **new** legal rules covered

$|\Gamma(a) \cap \Gamma(S)|$ rules already covered (redundancy)

Algorithm 1: ExpertGraphRAG retrieval for query x

1. **Route:** match x to concept cards and keep the top-8 concepts with weights w_r
2. **Support:** $g(a) = \max_r w_r \cdot B_{r,a}$; $\Gamma(a)$ = rules supporting a
3. **Pool:** dense top-50 $\cup$ graph top-30 articles
4. **Greedy:** $S \leftarrow S \cup \{\arg\max_a \Delta(a|S)\}$ until $|S| = 8$

Monotone submodular coverage gain $\rightarrow$ greedy approximation. $\lambda = 0.55 / 0.18 / 0.06$.

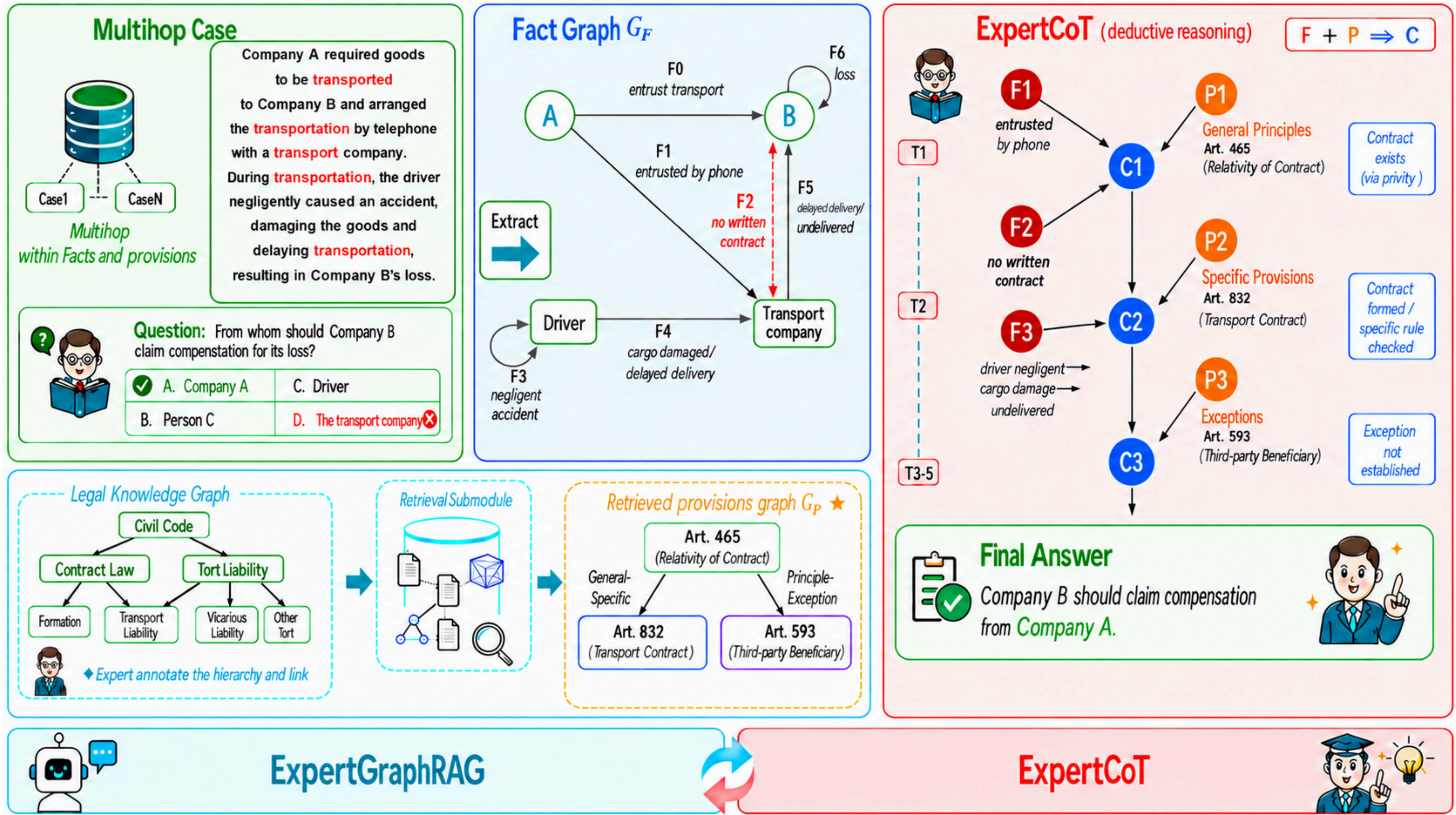
ExpertCoT

Reason as a legal syllogism

P · Provision	Major premise: controlling rules taken only from retrieved provisions, with priority, provisos, exceptions and remedies
F · Fact	Minor premise: legally relevant facts, with no legal conclusions yet
C · Conclusion	A judgment per option, checking subject, predicate, elements and legal effect

11 explicit instructions, refined through expert-in-the-loop error analysis, all in a single generation pass.

LEGO injects expert knowledge into both retrieval and reasoning

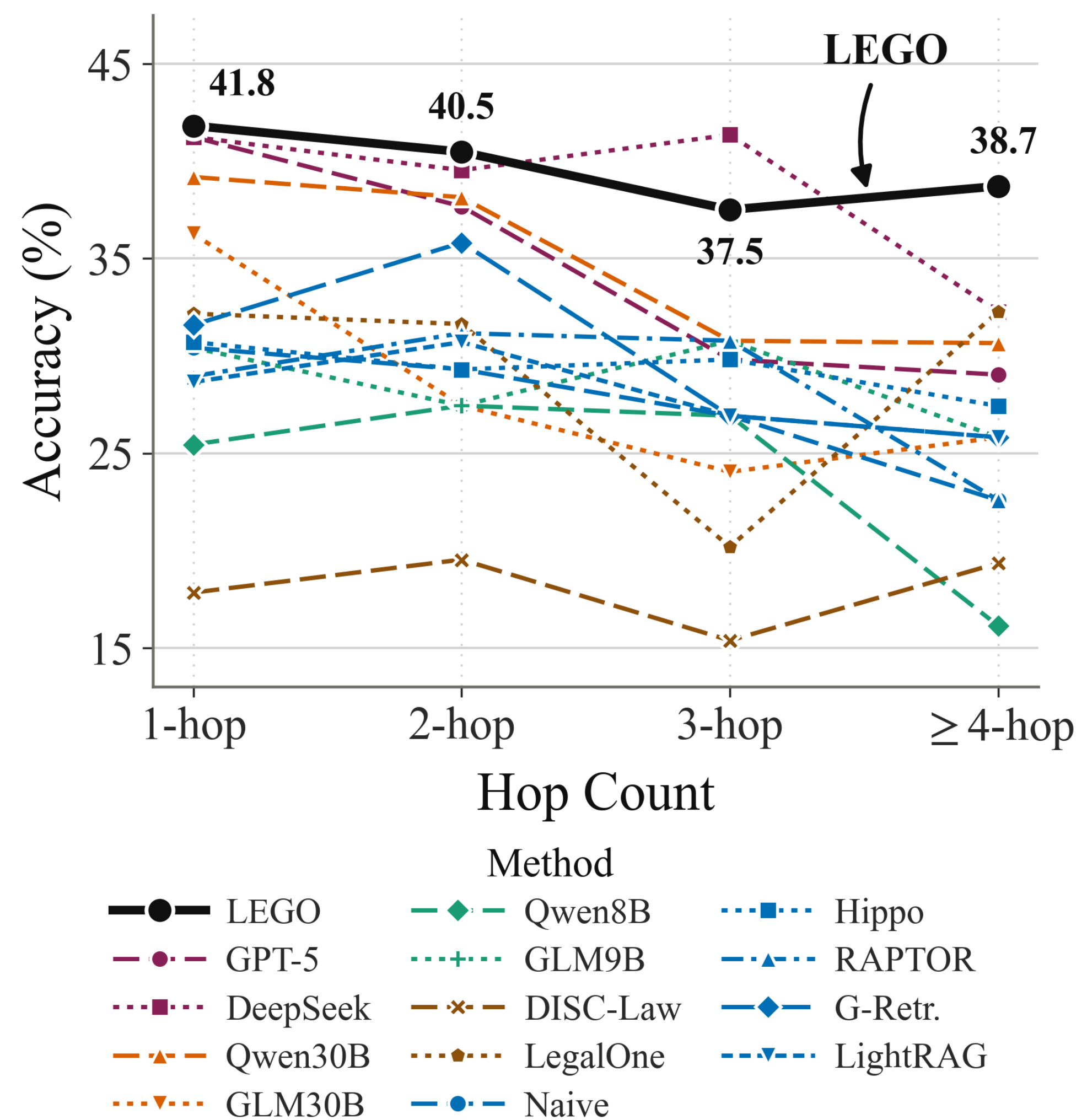


ExpertGraphRAG queries with the case facts, question and options, and returns an instance-specific provision subgraph G_P . **ExpertCoT** then combines provisions and facts into a P–F–C analysis. The Fact and Conclusion graphs are conceptual, not materialised.

LEGO (8B) beats strong RAG baselines and legal LLMs, comparable to much larger LLMs

Method	Size	All	1-hop	2-hop	3-hop	≥ 4
<i>Closed-source & large general LMs</i>						
GPT-5	API	37.48	<u>41.23</u>	37.67	29.81	29.03
DeepSeek-V3	671B	<u>39.97</u>	<u>41.23</u>	<u>39.53</u>	41.35	<u>32.26</u>
<i>Open larger LMs</i>						
Qwen3-30B-A3B	30B	36.93	39.18	38.14	30.77	30.65
GLM-4.7-Flash	30B	30.98	36.26	27.44	24.04	25.81
<i>Small open-source LMs</i>						
Qwen3-8B	8B	25.45	25.44	27.44	26.92	16.13
GLM-4-9B-chat	9B	29.18	30.41	27.44	30.77	25.81
<i>Legal-domain LMs</i>						
DISC-LawLLM	7B	18.12	17.84	19.53	15.38	19.35
LegalOne	8B	30.29	32.16	31.63	20.19	<u>32.26</u>
<i>RAG baselines (same Qwen3-8B reader and corpus)</i>						
Naive RAG	8B	28.91	30.41	29.30	26.92	22.58
HippoRAG 2	8B	29.88	30.70	29.30	29.81	27.42
RAPTOR	8B	29.32	28.95	31.16	30.77	22.58
G-Retriever	8B	31.67	31.58	35.81	26.92	25.81
LightRAG	8B	28.77	28.65	30.70	26.92	25.81
LEGO (ours)	8B	40.53	41.81	40.47	37.50	38.71

LawExamQA_Civil exact-match accuracy (%), 723 items (342 / 215 / 104 / 62 by hop count). **Bold** = best, underline = second best.

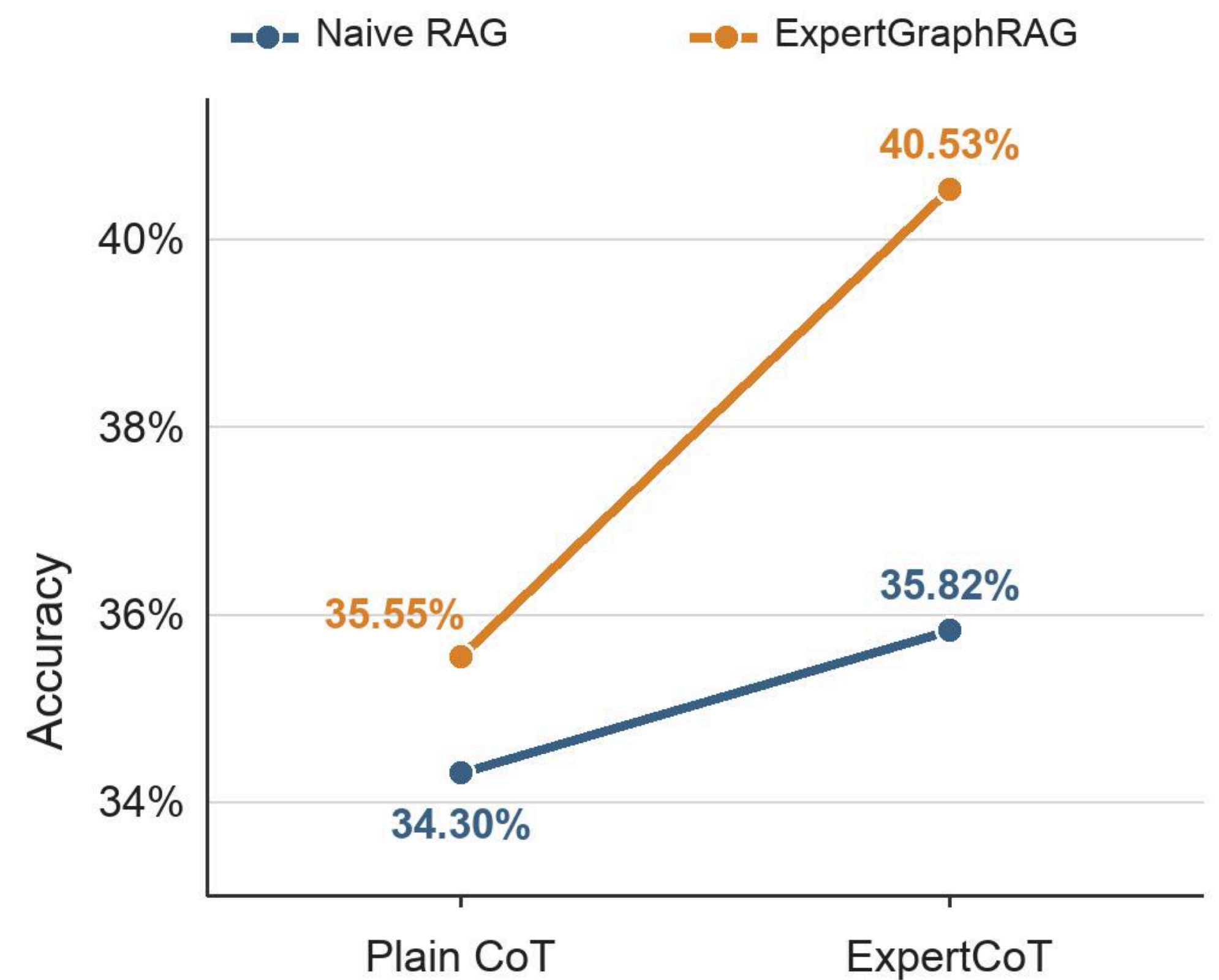


LEGO stays nearly flat across hops (41.8 $\rightarrow$ 38.7), while GPT-5 drops 12.2 pp from 1-hop to ≥ 4 -hop.

The two modules are complementary

Model	Configuration	Acc	F1	Δ Acc	Δ F1
<i>No-RAG baselines</i>					
Qwen3-8B	zero-shot	0.2545	0.5382	$\uparrow 0.1508$	$\uparrow 0.0305$
Qwen3-8B	CoT	0.2918	0.4728	$\uparrow 0.1135$	$\uparrow 0.0959$
Qwen3-8B	IRAC-CoT	0.3112	0.4973	$\uparrow 0.0941$	$\uparrow 0.0714$
<i>RAG + standard CoT</i>					
Qwen3-8B	Naive RAG + CoT	0.3430	0.5159	$\uparrow 0.0623$	$\uparrow 0.0528$
Qwen3-8B	Naive RAG + IRAC-CoT	0.3223	0.5312	$\uparrow 0.0830$	$\uparrow 0.0375$
Qwen3-8B	G-Retriever + CoT	0.3375	0.5074	$\uparrow 0.0678$	$\uparrow 0.0613$
Qwen3-8B	G-Retriever + IRAC-CoT	0.3347	0.5331	$\uparrow 0.0706$	$\uparrow 0.0356$
<i>RAG + ExpertCoT</i>					
Qwen3-8B	Naive RAG + ExpertCoT	0.3582	0.5280	$\uparrow 0.0471$	$\uparrow 0.0407$
Qwen3-8B	G-Retriever + ExpertCoT	0.3555	0.5415	$\uparrow 0.0498$	$\uparrow 0.0272$
<i>ExpertGraphRAG + standard CoT</i>					
LEGO	CoT	0.3555	0.5312	$\uparrow 0.0498$	$\uparrow 0.0375$
LEGO	IRAC-CoT	0.3734	0.5456	$\uparrow 0.0319$	$\uparrow 0.0231$
<i>ExpertGraphRAG + ExpertCoT</i>					
LEGO	Full system	0.4053	0.5687	–	–
<i>Gold article</i>					
Gold Article	+ CoT	0.3472	0.5257	–	–
Gold Article	+ IRAC-CoT	0.3790	0.5468	–	–
Gold Article	+ ExpertCoT	0.4122	0.5894	–	–

Δ = full LEGO minus the row. Darker shading means a larger gap.



Each module helps more when the other is present.

- **ExpertGraphRAG** adds +1.25 pp with plain CoT but **+4.71 pp** with ExpertCoT
- **ExpertCoT** adds +1.52 pp with Naive RAG but **+4.98 pp** with ExpertGraphRAG

ExpertGraph nearly closes the gold gap

Method	R@8	R@10	R@20	Acc	F1	GCR
Naive RAG	27.99	34.70	50.52	28.91	52.16	0.61
BM25 (Civil Code)	45.55	48.21	57.90	29.05	52.96	0.64
Dense (Civil Code)	68.36	71.93	79.19	29.60	55.07	0.73
LEGO w/ ExpertGraph	72.53	74.58	81.07	30.29	54.78	0.85
Gold Article	100	100	100	31.12	55.41	1.00

Recall@k (%) of gold provisions on LawExamQA_Civil; QA with a Qwen3-8B zero-shot reader on the top-8 provisions. GCR = share of the zero-shot $\rightarrow$ gold accuracy gap that is closed.